LOKE FU (He/Him) | HKPR

Hong Kong | +852-54883808 | fuyicheng20011119@gmail.com | GitHub: lokefu | LinkedIn: lokefu

SUMMARY

Results-driven Data Scientist specializing in Generative AI ecosystem, with expertise in architecting Agentic RAG workflows (LangGraph) and fine-tuning Open-source LLMs (Mistral, Llama). Proven track record in leveraging Multimodal LLMs (MLLM) and Chain-of-Thought (CoT) reasoning to solve complex visual and textual challenges. Skilled in optimizing LLM performance through Instruction-tuning, Prompt Engineering, and high-performance inference deployment.

PROFESSIONAL EXPERIENCE

Tictag Singapore (Hong Kong Office) — **Hong Kong**
Data Scientist — *Nov 2024 - Mar 2026*

- **Architected a production-grade RAG Agent** using **Python, LangGraph & LangChain** to orchestrate a modular AI state-machine; integrated Multi-Query Expansion and Rewrite, Hybrid Ensemble Retrieval (Jina, MPNet & BM25), and Flashrank Re-ranking to minimize hallucinations and boost retrieval precision in Mistral-7B responses from **Vector Database (FAISS)**.
- **Engineered an automated data-to-insight pipeline** by Few-Shot prompting **Vertex AI (Google Gemini)** and defined a JSON output schema, successfully automating structured text analysis output, boosting team productivity by 90%.
- **Advanced Long-Video Understanding** by implementing **Multimodal LLMs (GPT/Transformer architectures)** with a Chain-of-Thought (CoT) approach, enabling customer identification, event detection, video captioning and video Q&A.
- **Spearheaded computer vision features** for retail analytics using PyTorch, including people detection & tracking, group detection, event classification and multi-camera 2D projection to provide comprehensive behavioral data for LLM analysis.
- **Developed a high-accuracy product recognition system** (96% accuracy) by integrating object detection and segmentation with **MySQL** for metadata management and an image retrieval embedding database.

Centre for Advances in Reliability & Safety (HK PolyU & University of Maryland) — **Hong Kong**
Data Scientist — *Feb 2024 - Nov 2024*

- **Awarded a Gold Medal** at the 49th International Exhibition of Inventions Geneva for the "Cable Prognosis" project, utilizing **Python and LSTMs** for sequential data modeling and predictive reliability for Remaining Useful Life (RUL).
- **Built and optimized** deep learning (CNN-based), supervised (XGBoost/MLP) and unsupervised (VAE/DBSCAN) machine learning models for industrial fault detection, achieving **99% accuracy** by integrating physical simulation data via **BigQuery**.
- **Collaborated** with senior engineers to streamline UI/UX development for commercial-grade applications.

The University of Hong Kong (HKU), School of Computing — **Hong Kong**
Research Assistant (Master's Researcher) — *Sep 2022 - Jan 2024*

- **Designed and developed** *DreamMoRe*, a novel subject-driven Text-to-Image framework using **PyTorch** that enhances DreamBooth through Text Rewriting, Image Augmentation, and Multi-Modal Loss integration.
- **Executed Supervised Fine-Tuning (SFT, LoRA)** using Hugging Face Transformers and **Instruction-tuning** on LLMs to enhance complex prompt comprehension and cross-modal alignment.
- Utilized **CLIP-based semantic control mechanisms** to reinforce prompt accuracy and subject fidelity in generation.
- Established comprehensive evaluation benchmarks using both **quantitative (CLIP score)** and **qualitative (Human-in-the-loop)** metrics to validate model alignment.

EDUCATION

The University of Hong Kong (HKU) — *2022 - 2024*
Master of Science in Artificial Intelligence

The University of Hong Kong (HKU) — *2018 - 2022*
Bachelor of Science in Statistical Decision Sciences (Decision Analytics)

- **Academic Award:** HKU Foundation Entrance Scholarships (Full 4-year scholarship)
- **Exchange Program:** Harvard University, Massachusetts, U.S. (2019)

TECHNICAL SKILLS & LANGUAGES

Programming: Python, R
LLM/Generative AI: LangChain, LangGraph, Ollama, Hugging Face Transformers, LlamaIndex, FAISS, Vertex AI
ML/AI: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib
Data & Dev Tools: SQL (MySQL / BigQuery), Git, Docker, AWS, GCP, FastAPI, Gradio, HTML/CSS, JavaScript
Languages: English (Proficient), Mandarin (Native), Cantonese (Conversational)